Screen plan & navigation

Account-based access, shared group records and simple personal profiles.

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Updated scope: Every member has an account. Shared groups, expenses and settlements will use Firestore; profiles hold full name, optional photo and optional IBAN. Balances are derived from expenses and settlements. This replaces the single-device scope and is proposed for TA review.

MAIN PATH

S01 Log in → **S03 My groups** → **S05 Overview**

ACTION / SOURCE	DESTINATION / RETURN
Create account / S01	S02 -> valid confirmation -> S03
Create group / S03	S04 -> S14 invite code -> S05
Join group / S03	S13 -> preview + confirm -> S05
Profile / S03	S11 -> S12 edit -> S11; Back -> S03
Add expense / S05	S06 -> confirmation -> S05
View or edit / S05	S07 -> S06 edit -> S07; Back -> S05
Group tabs / S05, S08, S10	Expenses S05 <-> Balances S08 <-> Members S10
Payment / S08	S09 -> confirmation -> S08
Owner invite / S10	S14 -> Done / Back -> S05

Stage boundary: Step 2 shows mock records and simulated login, profile, invite and save states. Step 3 connects Firebase Authentication, Firestore and Cloud Storage for optional photos. Provider will manage UI state. No Firebase connection or live synchronization is claimed in these wireframes.

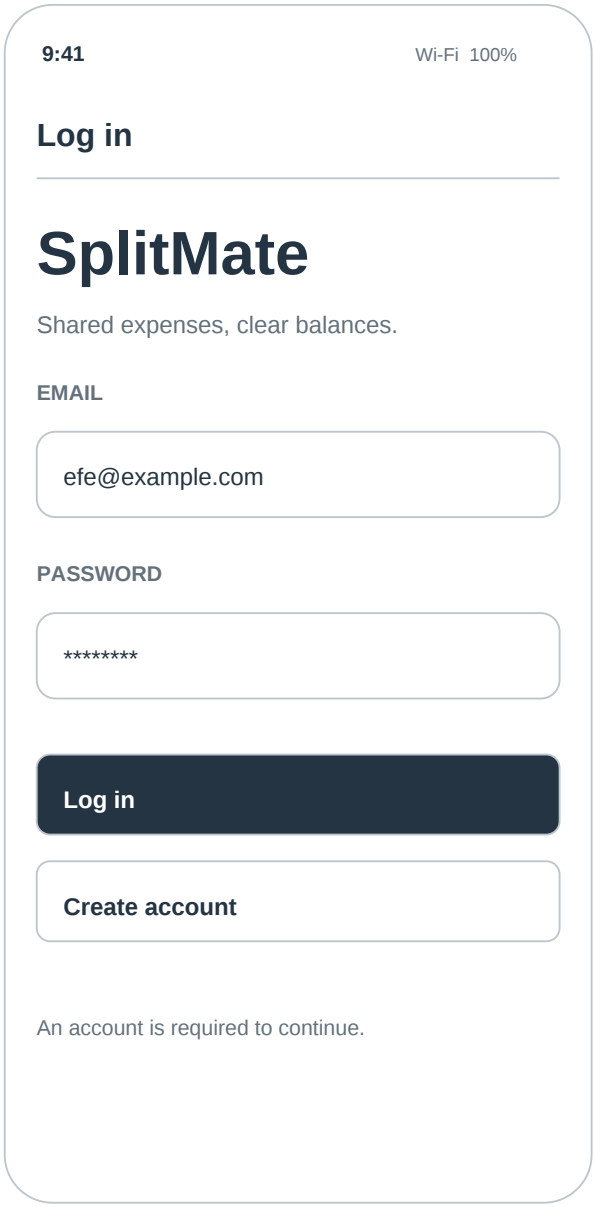
Legend: Dark buttons are primary actions; light cards are selectable records. Each screen has a description and exact destinations. Group tabs retain the selected group. Photo/IBAN are optional. Step 2 coordinator selection and TA feedback remain pending.

Repository: https://github.com/efe-akkus/cs310-project

Entry & account screens

Two portrait wireframes per page. Labels and numbers are illustrative.

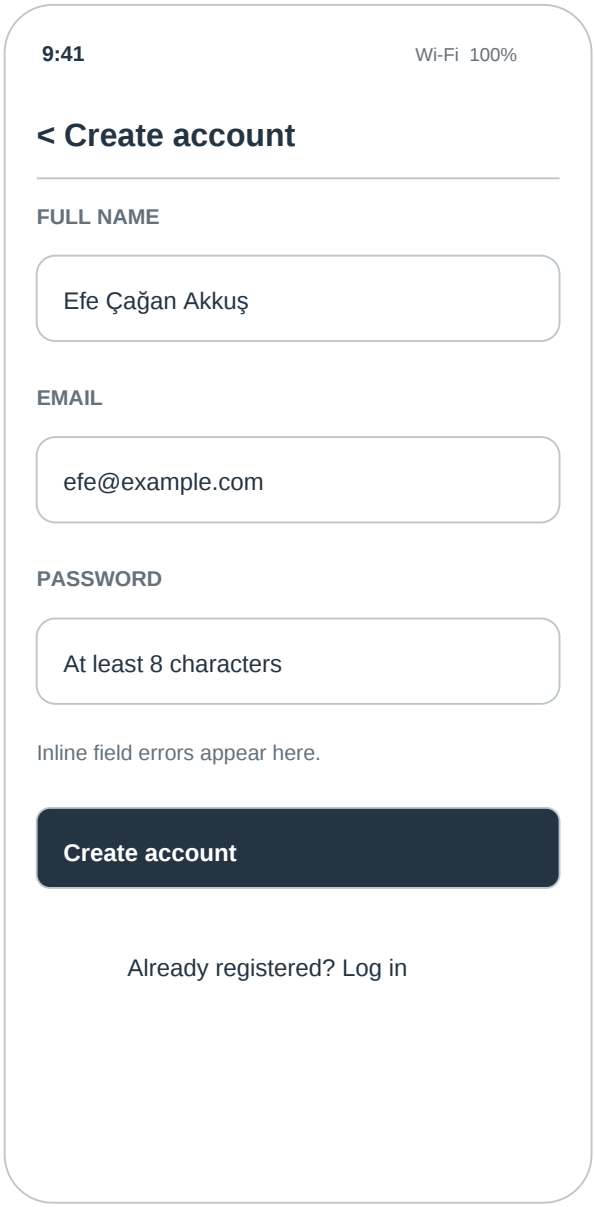
01 Log in



Sign in with email and password to view your shared groups. An account is required; guest access is removed. Step 2 uses simulated success/error states; Firebase Authentication follows in Step 3.

Navigation: Log in -> S03; Create account -> S02.

02 Create account



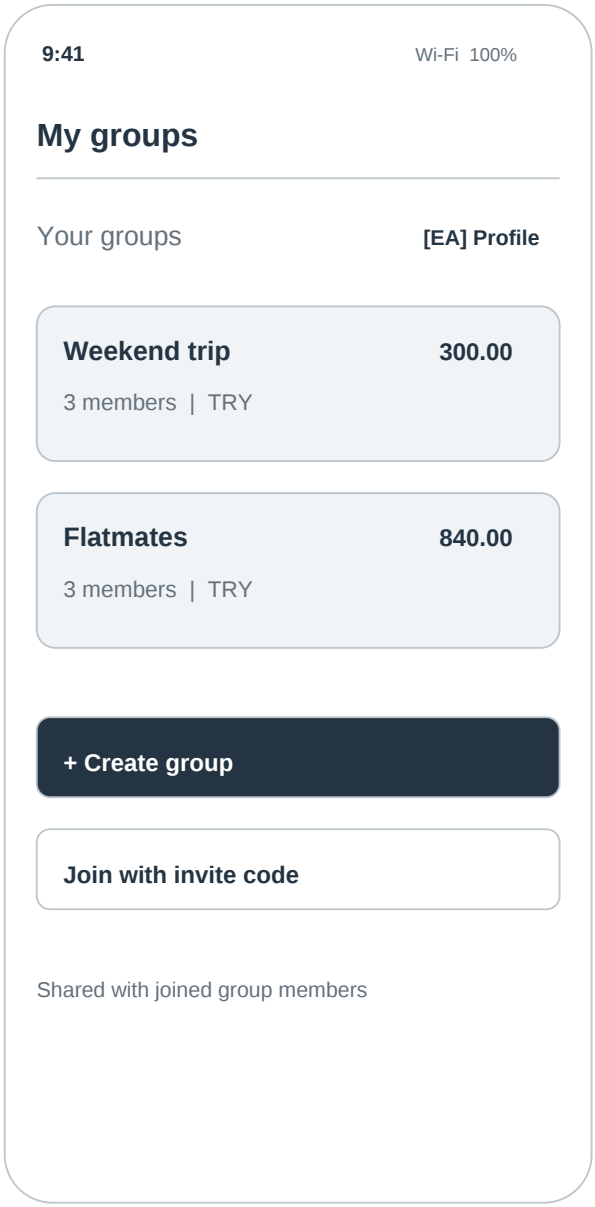
Enter full name, email and password. Confirm valid details to enter S03. Photo and IBAN are optional and added later in S12. Authentication is simulated in Step 2.

Navigation: Create account -> confirmation -> S03; Back -> S01.

Groups & creation

Two portrait wireframes per page. Labels and numbers are illustrative.

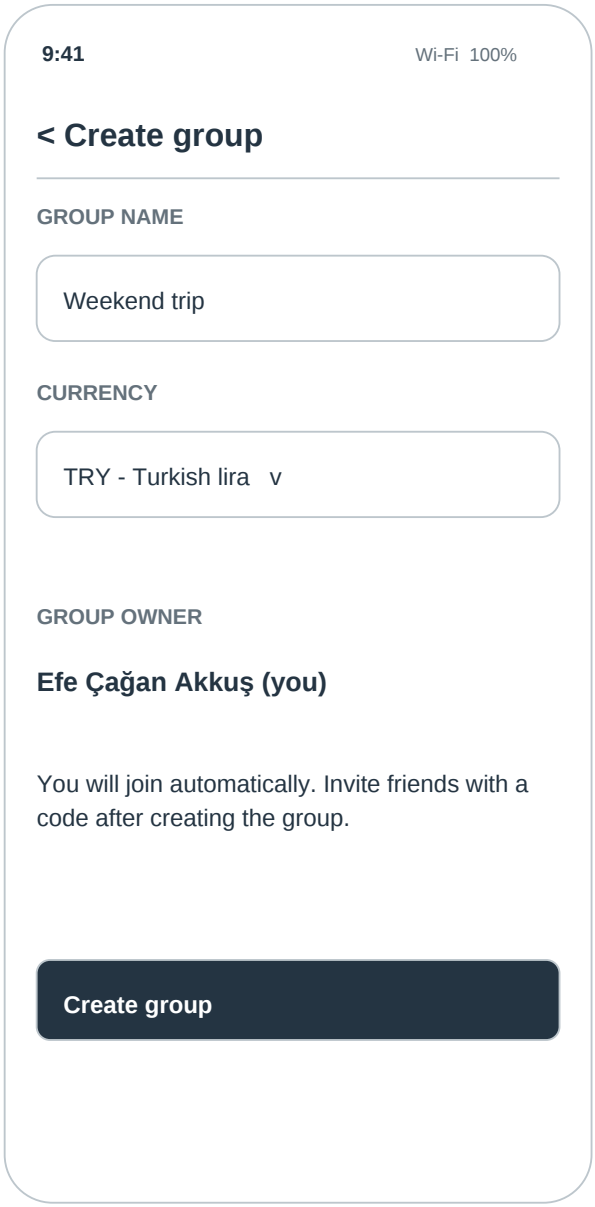
03 My groups



Home lists only groups joined by the signed-in user. Create a group or join with an invite code. The profile button opens your own profile. Shared records will sync through Firestore in Step 3.

Navigation: Group -> S05; Create -> S04; Join -> S13; Profile -> S11.

04 Create group



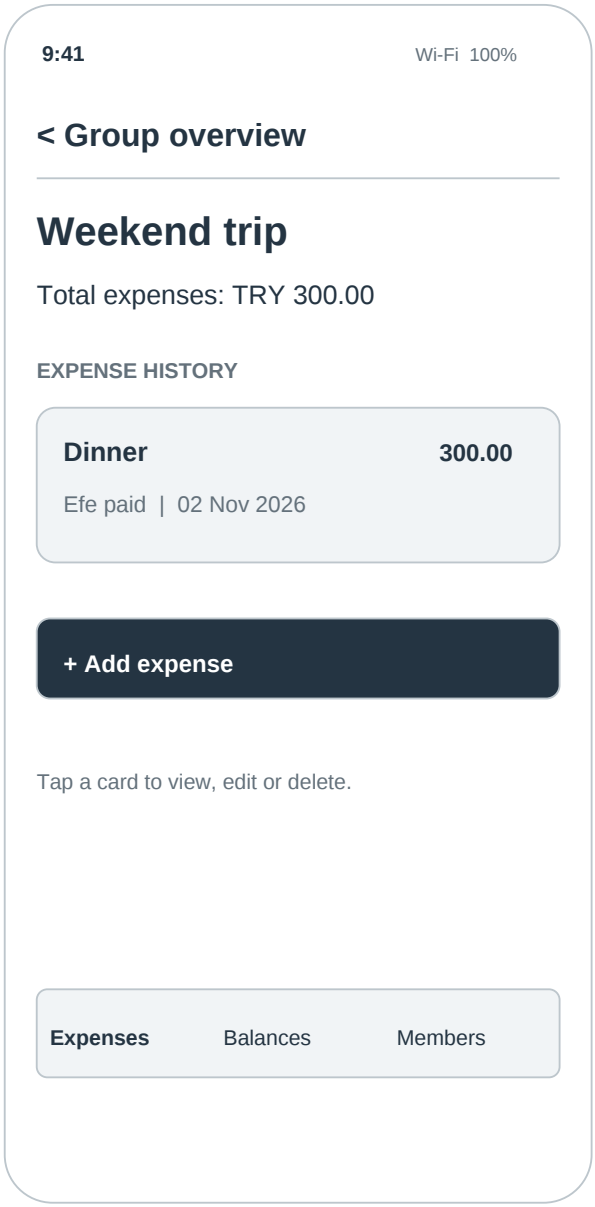
Choose the group name and currency. The creator becomes its first member and owner. Do not enter other people as local labels; invite registered accounts after creation.

Navigation: Create -> confirmation -> S14; Cancel / Back -> S03.

Expense history & entry

Two portrait wireframes per page. Labels and numbers are illustrative.

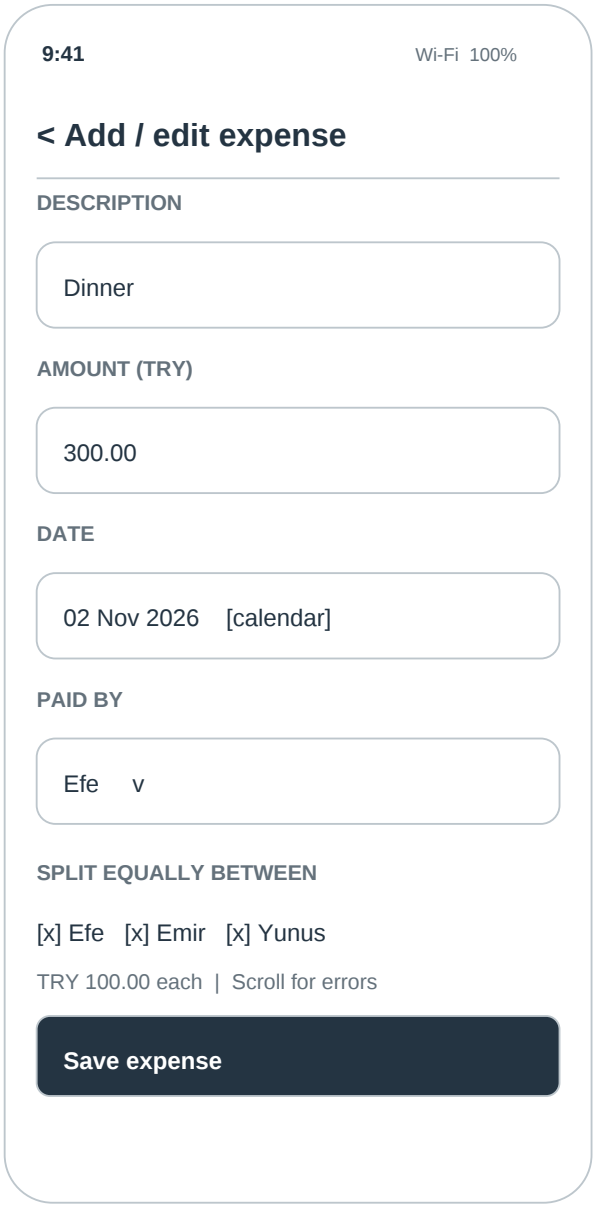
05 Group overview



View the group total and expense history. Expense cards show payer, amount and date. Open a card to inspect it; use the tabs for balances or members.

Navigation: Add expense -> S06; Expense -> S07; Balances -> S08; Members -> S10.

06 Add / edit expense



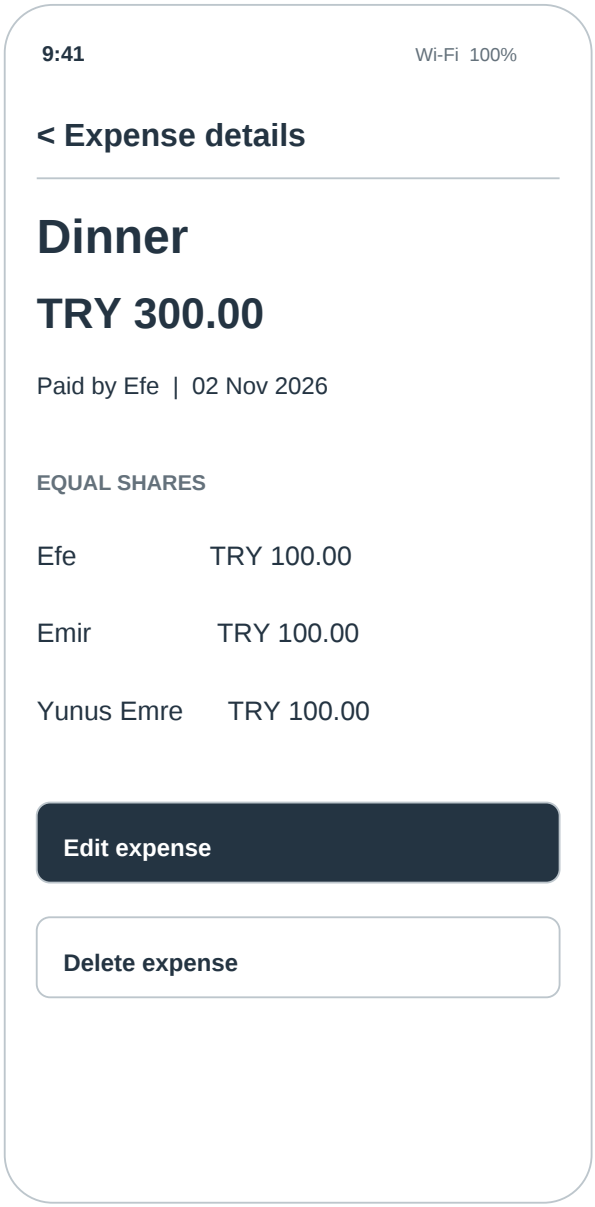
Enter description, positive amount, date, payer and participants selected from joined members. Split equally. Editing pre-fills the form. Only the expense creator can edit/delete it; others can view.

Navigation: Valid Save -> confirmation -> S05 (add) or S07 (edit); Cancel -> source.

Expense details & balances

Two portrait wireframes per page. Labels and numbers are illustrative.

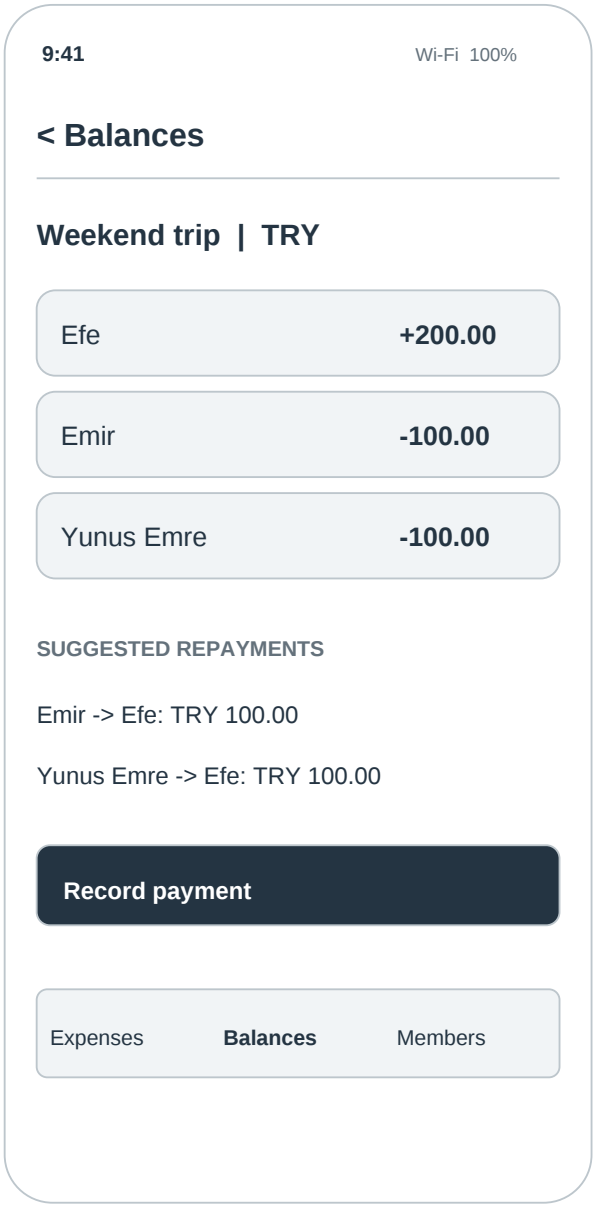
07 Expense details



Inspect the payer, participants and exact shares. For the expense creator, show Edit and Delete; other members see read-only details. Confirming deletion refreshes shared history and balances.

Navigation: Edit -> S06; Delete -> confirmation -> S05; Back -> S05.

08 Balances



Show net balances and suggested repayments. Positive means the member should receive money; negative means they owe money. The group ledger remains balanced.

Navigation: Record payment -> S09; Expenses -> S05; Members -> S10.

Payments & members

Two portrait wireframes per page. Labels and numbers are illustrative.

09 Record payment

9:41 Wi-Fi 100%

< Record payment

EXAMPLE: EMIR IS SIGNED IN
FROM (YOU)

Emir Keskin

TO

Efe Çağan Akkuş v

RECIPIENT IBAN (ILLUSTRATIVE)

TR** **** **** ****
**** **** 1234 **Copy IBAN**

AMOUNT (TRY)

40.00

Outstanding: TRY 100.00

Records only. No bank transfer.

Record payment

Sender is the signed-in user; select a suggested recipient. Show their optional IBAN with Copy, or an unavailable message. Record a payment already made outside the app; confirm to update balances.

Navigation: Valid Record -> confirmation -> S08; Cancel / Back -> S08.

10 Group members

9:41 Wi-Fi 100%

< Group members

Weekend trip | 3 members

[EA] Efe Çağan Akkuş
Owner (you)

[EK] Emir Keskin
Member

[YY] Yunus Emre Yolal
Member

Invite members (owner)

Profile changes: My groups > Profile

Expenses Balances **Members**

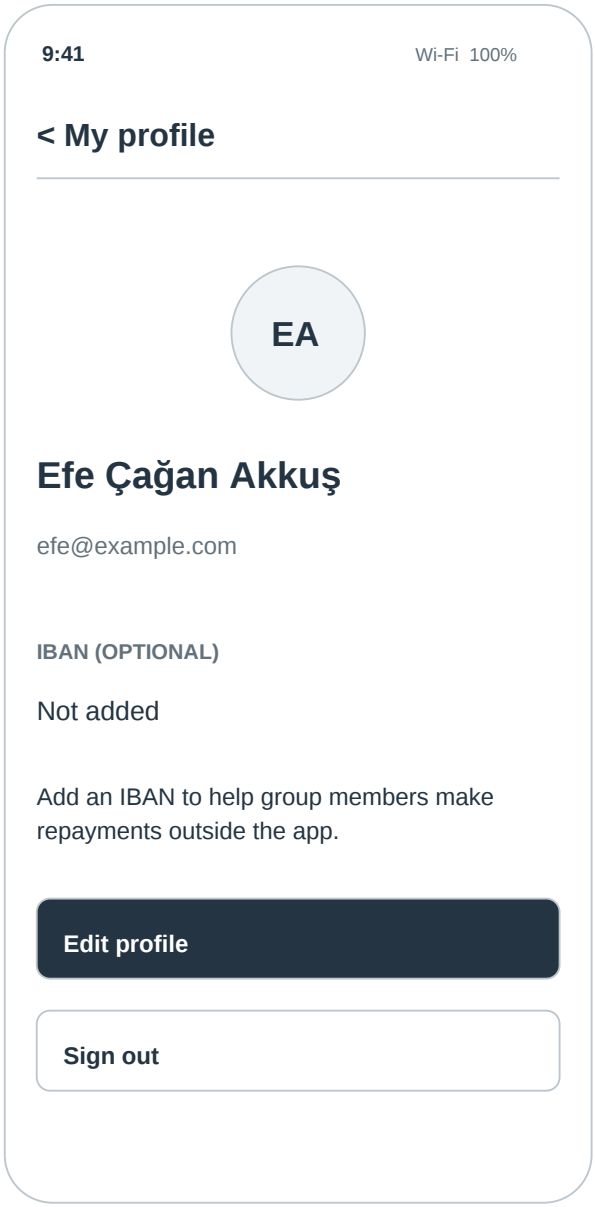
List joined accounts and their profile photos or initials. Members edit only their own profile. The owner manages invite codes; all group members may view the list. IBANs are not shown here.

Navigation: Owner: Invite members -> S14; Expenses -> S05; Balances -> S08.

Personal profile

Two portrait wireframes per page. Labels and numbers are illustrative.

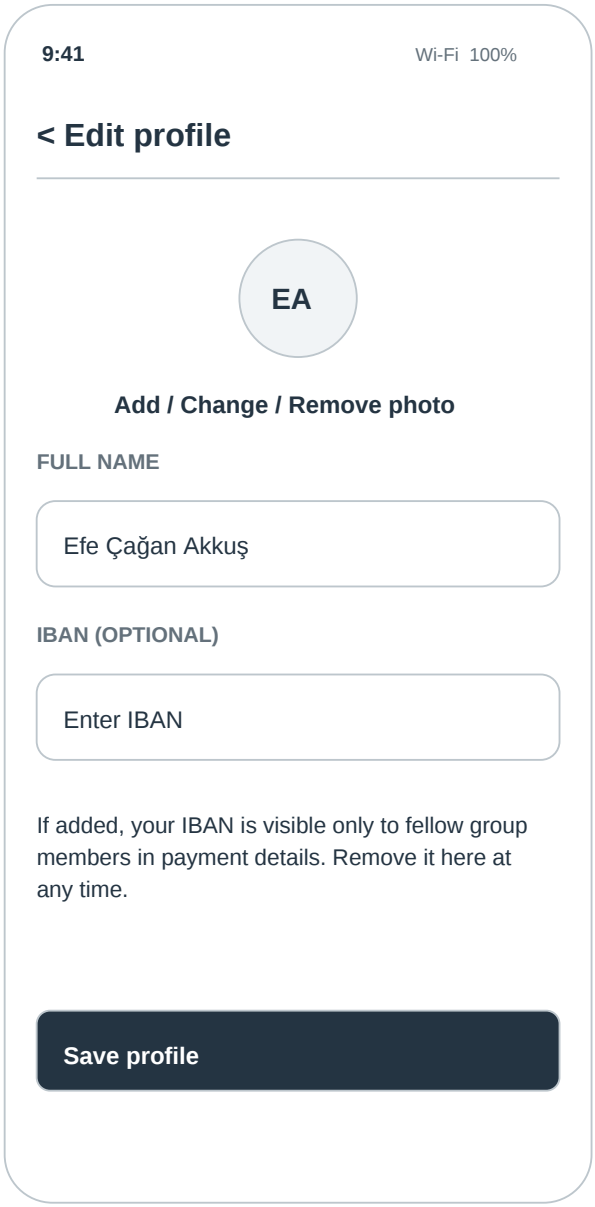
11 My profile



View your own full name, photo or initials, email and optional IBAN. Email is read-only for this scope. Sign out clears the signed-in session and returns to login.

Navigation: Edit profile -> S12; Sign out -> confirm -> S01; Back -> S03.

12 Edit profile



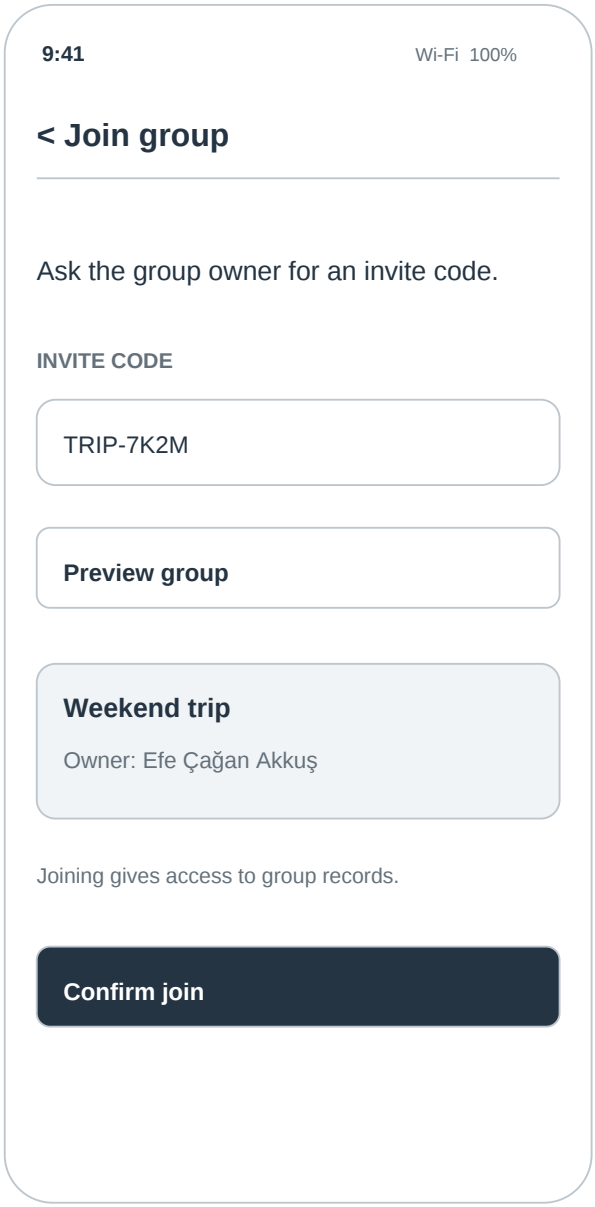
Full name is required. Add or remove an optional photo and IBAN. A photo permission denial leaves initials. Explain IBAN visibility to fellow group members in payment details before saving.

Navigation: Save -> confirmation -> S11; Cancel / Back -> S11.

Joining & inviting

Two portrait wireframes per page. Labels and numbers are illustrative.

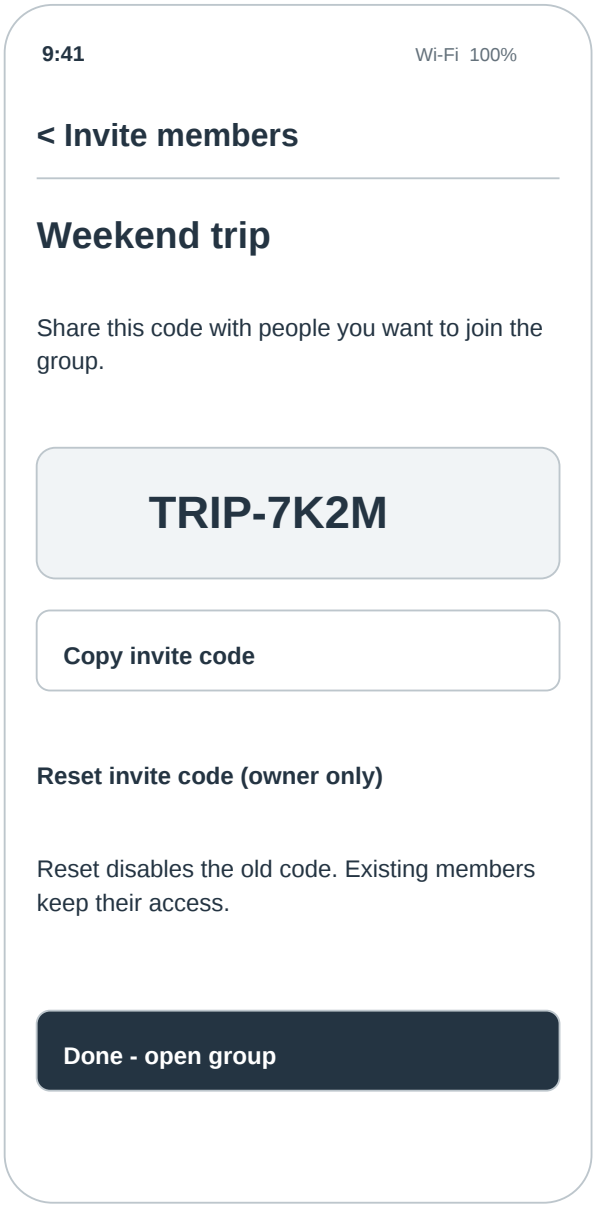
13 Join group



Enter an invite code, preview the group name and owner, then confirm joining. Invalid or expired codes show an inline error. Already joined groups open directly without a duplicate membership.

Navigation: Preview -> same screen; Confirm join / Open group -> S05; Back -> S03.

14 Invite members



Owner-only screen shows the invite code with Copy. Share it outside the app. Reset invalidates the old code after confirmation. Non-owners cannot access this screen.

Navigation: Copy -> stays; Reset -> confirmation; Done / Back -> S05.

Interaction states & review checklist

Supporting states keep the main screens clear and the core workflow complete.

Account and group access

Sign-in is required. Members see only their joined groups. Group creators own invitations; invite codes are validated before joining. A reset code stops future joins but does not remove current members. Duplicate joins open the existing group.

Forms and optional profile fields

Required: full name, valid email and password on sign-up; group name and currency; expense description, positive amount, date, payer and participants. IBAN may be blank; if entered, validate its format and checksum. Photo and IBAN never block account creation.

Photo and IBAN visibility

A denied photo permission or failed photo upload keeps initials and allows retry. IBAN is absent from group lists/member cards; show it only to fellow members in recipient payment details. Copy uses the full stored IBAN and shows a copied message. Missing IBAN: show "IBAN not added"; payments can still be recorded.

Confirmation and deletion

Valid forms show a summary with Cancel / Confirm before saving. S07 shows Edit/Delete only to the expense creator. Delete confirmation explains that balances will change. Cancel preserves the record. Other members can view but cannot edit that expense.

Balances and settlement records

Derive balances from expense shares and settlements. Example: Efe +200, Emir -100, Yunus Emre -100. Emir records a TRY 40 payment to Efe: +160, -60, -100. Sender is the current user; recipient differs, amount is positive and cannot exceed the suggested repayment. No real bank transfer occurs.

Empty, loading and failure states

Empty groups: Create / Join. Empty history: Add expense. Zero balances: All settled. Loading disables repeat submissions. In Step 3, network/auth/save errors show Retry and preserve input; do not show success before server confirmation. Step 2 simulates these states with mock data.

Implementation and review

Step 2.2 will use model-backed expense Cards with remove, reusable widgets, a custom font, bundled image and network image with fallback. Use scrollable, keyboard-aware layouts on portrait phones. Obtain TA review of the expanded shared-group scope before Flutter UI implementation.

Planned data responsibilities (Step 3)

Authentication: accounts. Firestore: profiles, memberships, groups, expenses and settlements. Storage: optional photos. Enforce member-only group access and owner-only profile editing with backend rules; screen visibility alone is insufficient.